

Disproof of Conjecture 00000004310: imaginary quadratic fields of the same discriminant have isomorphic unit groups

jilint777

2026-10-04

Abstract

Conjecture 00000004310 asserts that there exist two imaginary quadratic fields of the same discriminant whose class groups have isomorphic 3-parts while their unit groups are not isomorphic, and that the smallest such discriminant has seven digits. This is false, whatever the class-group condition means. The discriminant determines an imaginary quadratic field: every such field is $\mathbb{Q}(\sqrt{m})$ for a unique squarefree $m < 0$, its discriminant is m or $4m$, and $m \mapsto \text{disc } m$ is injective. So two fields with the same discriminant are equal, and their unit groups are equal. A second, independent argument: the unit group of an imaginary quadratic field, and more generally of an imaginary quadratic order, is cyclic of order $w(D)$, where $w(-4) = 4$, $w(-3) = 6$ and $w(D) = 2$ otherwise, so it is a function of the discriminant D alone. There is therefore no witness discriminant at all, and in particular no smallest one with seven digits. The Lean 4 development (core library only) defines squarefree integers, the discriminant, the rings of integers $\mathcal{O}_m$ and the quadratic orders of every discriminant as rings of pairs, their unit groups and isomorphisms of unit groups from scratch. It proves injectivity of the discriminant, the complete list of units, and the negation of the claim, for an arbitrary class-group predicate. A standard-library Python script checks the same facts numerically by other methods, including the reading with binary quadratic forms.

1 The statement

The conjecture reads:

Definition: The p -part of the class group and the unit group are two invariants of fields of the same discriminant. Conjecture: There exist two imaginary quadratic fields of the same discriminant whose 3-parts of the class groups are isomorphic while their unit groups are not, and the smallest discriminant is a seven-digit value.

The Chinese version says the same: there exist two imaginary quadratic fields with the same discriminant whose class groups have isomorphic 3-parts and whose unit groups are not isomorphic, and the minimal discriminant is a seven-digit number.

The conjecture is a conjunction of two clauses:

- (E) there are imaginary quadratic fields K_1, K_2 with $D_{K_1} = D_{K_2}$, $\text{Cl}(K_1)[3^\infty] \cong \text{Cl}(K_2)[3^\infty]$ and $\mathcal{O}_{K_1}^\times \not\cong \mathcal{O}_{K_2}^\times$;
- (M) the smallest $|D|$ for which (E) holds has seven digits.

We show that (E) is false. Then (M) is false as well, since there is no witness discriminant at all. Refuting (E) suffices to refute the conjecture.

2 Background

We use the following standard facts; see D. A. Marcus, *Number Fields*, 2nd ed., Springer 2018, Chapter 2 (quadratic fields, their rings of integers and discriminants), and D. A. Cox, *Primes of the Form $x^2 + ny^2$* , 2nd ed., Wiley 2013, §2 (forms) and §7 (orders).

- (F1) *Classification.* Every quadratic field is $\mathbb{Q}(\sqrt{m})$ for a unique squarefree integer $m \notin \{0, 1\}$. It is imaginary iff $m < 0$. Uniqueness: $\mathbb{Q}(\sqrt{m}) = \mathbb{Q}(\sqrt{m'})$ forces m/m' to be a square in $\mathbb{Q}$, hence $m = m'$ for squarefree m, m' . This holds for subfields of $\mathbb{C}$ and for abstract fields up to isomorphism.
- (F2) *Ring of integers.* For squarefree m , $\mathcal{O}_K = \mathbb{Z}[\sqrt{m}]$ if $m \equiv 2, 3 \pmod{4}$ and $\mathcal{O}_K = \mathbb{Z}[\frac{1+\sqrt{m}}{2}]$ if $m \equiv 1 \pmod{4}$.
- (F3) *Discriminant.* $D_K = 4m$ if $m \equiv 2, 3 \pmod{4}$ and $D_K = m$ if $m \equiv 1 \pmod{4}$.
- (F4) *Orders.* For every $D < 0$ with $D \equiv 0, 1 \pmod{4}$ there is exactly one imaginary quadratic order of discriminant D up to isomorphism, namely $\mathcal{O}_D = \mathbb{Z}[\frac{D+\sqrt{D}}{2}]$. Every order in an imaginary quadratic field is of this form (Cox, §7.A: an order of conductor f is $\mathbb{Z} + f\mathcal{O}_K$, of discriminant $f^2 D_K$).

3 The discriminant determines the field

For an integer m put $\text{disc } m = m$ if $m \equiv 1 \pmod{4}$ and $\text{disc } m = 4m$ otherwise.

Lemma 1. *The map $m \mapsto \text{disc } m$ is injective on $\mathbb{Z}$. On squarefree m its values satisfy $\text{disc } m \equiv 1 \pmod{4}$ or $\text{disc } m \equiv 8, 12 \pmod{16}$, and m is recovered as $m = \text{disc } m$ if $\text{disc } m \equiv 1 \pmod{4}$ and $m = \text{disc } m/4$ otherwise.*

Proof. Suppose $\text{disc } m_1 = \text{disc } m_2$. If $m_1 \equiv m_2 \equiv 1 \pmod{4}$, then $m_1 = m_2$ directly. If neither is $\equiv 1$, then $4m_1 = 4m_2$. If $m_1 \equiv 1$ and $m_2 \not\equiv 1$, then $m_1 = 4m_2 \equiv 0 \pmod{4}$, a contradiction; the remaining case is symmetric. For squarefree m we have $4 \nmid m$. So if $m \not\equiv 1 \pmod{4}$ then $m \equiv 2, 3 \pmod{4}$ and $4m \equiv 8, 12 \pmod{16}$. The recovery formula follows because $4m \equiv 0 \not\equiv 1 \pmod{4}$. $\square$

Theorem 2. *If K_1, K_2 are imaginary quadratic fields with $D_{K_1} = D_{K_2}$, then $K_1 \cong K_2$. If they are subfields of $\mathbb{C}$, then $K_1 = K_2$. In either case $\mathcal{O}_{K_1}^\times \cong \mathcal{O}_{K_2}^\times$.*

Proof. By (F1) write $K_i \cong \mathbb{Q}(\sqrt{m_i})$ with $m_i < 0$ squarefree. By (F3), $\text{disc } m_1 = D_{K_1} = D_{K_2} = \text{disc } m_2$, so $m_1 = m_2$ by Lemma 1. Hence $K_1 \cong \mathbb{Q}(\sqrt{m_1}) \cong K_2$. Inside $\mathbb{C}$, an imaginary quadratic field is determined by the square class of m , so $K_1 = K_2$. A field isomorphism maps the ring of integers onto the ring of integers, so it restricts to an isomorphism of unit groups. $\square$

4 The unit groups depend only on the discriminant

This section gives a second argument that does not use Lemma 1, and it also covers orders.

Quadratic rings of pairs. For integers t, n let $R_{t,n} = \mathbb{Z}[\omega]$, $\omega^2 = t\omega - n$. Its elements are $a + b\omega$, $a, b \in \mathbb{Z}$, with

$$(a + b\omega)(c + d\omega) = (ac - nbd) + (ad + bc + tbd)\omega.$$

The ring has the conjugation $\overline{a + b\omega} = (a + tb) - b\omega$, the trace $\text{Tr}(a + b\omega) = 2a + tb$, and the norm $N(a + b\omega) = (a + b\omega)\overline{(a + b\omega)} = a^2 + tab + nb^2$. The norm is multiplicative and satisfies

$$4N(a + b\omega) = (2a + tb)^2 + (4n - t^2)b^2. \quad (1)$$

The discriminant of the basis $(1, \omega)$ is $\det(\text{Tr}(e_i e_j)) = 2(t^2 - 2n) - t^2 = t^2 - 4n$.

Orders and maximal orders. For $D \equiv 0, 1 \pmod{4}$ put $t = D \bmod 2 \in \{0, 1\}$ and $n = (t - D)/4$. Then $t^2 - 4n = D$, and $\omega = (t + \sqrt{D})/2$ satisfies $\omega^2 = t\omega - n$, so $R_{t,n} \cong \mathcal{O}_D$ is the order of discriminant D in (F4). For squarefree m , the parameters of (F2) are $(t, n) = (1, (1 - m)/4)$ when $m \equiv 1 \pmod{4}$, with $\omega = (1 + \sqrt{m})/2$, and $(t, n) = (0, -m)$ otherwise, with $\omega = \sqrt{m}$. These are exactly the parameters attached to $D = \text{disc } m$, so $\mathcal{O}_{\mathbb{Q}(\sqrt{m})} = \mathcal{O}_{\text{disc } m}$.

The map $a + b\omega \mapsto (x, y)$, where $x + y\sqrt{m} = 2(a + b\omega)$, identifies $R_{t,n}$ with the elements $(x + y\sqrt{m})/2$ of $\mathbb{Q}(\sqrt{m})$ and respects multiplication. Formula (F2) is the following statement.

Lemma 3 (integrality). *Let $m \not\equiv 0 \pmod{4}$ and $x, y \in \mathbb{Z}$. The element $(x + y\sqrt{m})/2$, which has trace $x \in \mathbb{Z}$, has norm $(x^2 - my^2)/4 \in \mathbb{Z}$ iff it lies in $\mathcal{O}_m$; that is, iff $x \equiv y \pmod{2}$ when $m \equiv 1 \pmod{4}$, and iff x, y are both even otherwise.*

Proof. A square z^2 is $\equiv 0 \pmod{4}$ if z is even and $\equiv 1 \pmod{4}$ if z is odd, so $x^2 - my^2 \equiv [x \text{ odd}] - m[y \text{ odd}] \pmod{4}$. If y is even this is $\equiv 0$ iff x is even. If y is odd it is $\equiv 1 - m$ or $-m$, according to the parity of x . Since $4 \nmid m$, the value is 0 exactly when x is odd and $m \equiv 1 \pmod{4}$. $\square$

Every algebraic integer $\alpha = u + v\sqrt{m} \in \mathbb{Q}(\sqrt{m})$ has this form: $2u = \text{Tr } \alpha \in \mathbb{Z}$, and $(2u)^2 - m(2v)^2 = 4N\alpha \in \mathbb{Z}$, so $m(2v)^2 \in \mathbb{Z}$, and $2v \in \mathbb{Z}$ because m is squarefree. Together with Lemma 3 this is the proof of (F2).

Theorem 4 (units). *Let $D < 0$, $D \equiv 0, 1 \pmod{4}$, and let (t, n) be as above. The units of $\mathcal{O}_D = R_{t,n}$ are the elements of norm 1. They are*

- $\pm 1, \pm\omega$ for $D = -4$ (here $\omega = i$),
- $\pm 1, \pm\omega, \pm(\omega - 1)$ for $D = -3$ (here $\omega = (1 + \sqrt{-3})/2$, $\omega^2 = \omega - 1$),
- ± 1 for every other D .

In each case $\mathcal{O}_D^\times$ is cyclic of order $w(D)$, where $w(-4) = 4$, $w(-3) = 6$ and $w(D) = 2$ otherwise. It is generated by ω , by ω , and by -1 , respectively.

Proof. Since $4n - t^2 = -D > 0$, (1) shows that $N \geq 0$. If $xy = 1$, then $N(x)N(y) = 1$ forces $N(x) = 1$. Conversely, if $Nx = 1$ then $x\bar{x} = 1$. Now let $N(a + b\omega) = 1$. By (1), $4 = (2a + tb)^2 + |D|b^2$ with $|D| \geq 3$, so $|b| \leq 1$ and $|2a + tb| \leq 2$, hence $|a| \leq 1$. If $|D| > 4$ then $b = 0$ and $a = \pm 1$. For $D = -4$ the equation is $a^2 + b^2 = 1$, and for $D = -3$ it is $a^2 + ab + b^2 = 1$. Enumerating the nine pairs (a, b) gives the lists. The powers of ω for $D = -3$ are $1, \omega, \omega - 1, -1, -\omega, 1 - \omega$, and $\omega^6 = 1$. $\square$

Corollary 5. *For imaginary quadratic fields, $\mathcal{O}_{\mathbb{Q}(\sqrt{m})}^\times \cong \mathbb{Z}/w(\text{disc } m)$. If $w(D_1) = w(D_2)$, then the unit groups of the orders of discriminants D_1, D_2 are isomorphic. In particular, equal discriminants give isomorphic unit groups. For orders of a fixed discriminant this holds without using Lemma 1.*

Theorem 6. *Clause (E) is false. More precisely, for every predicate $C(K_1, K_2)$ standing in for “ $\text{Cl}(K_1)[3^\infty] \cong \text{Cl}(K_2)[3^\infty]$ ”, there are no imaginary quadratic fields with $D_{K_1} = D_{K_2}$, $C(K_1, K_2)$ and $\mathcal{O}_{K_1}^\times \not\cong \mathcal{O}_{K_2}^\times$. Consequently there is no witness discriminant, and clause (M) is false.*

Proof. This follows from Theorem 2, or independently from Corollary 5. □

5 Readings of the statement

1. **“Two fields” as subfields of $\mathbb{C}$ or as fields up to isomorphism.** In both cases Theorem 2 gives $K_1 \cong K_2$, hence isomorphic unit groups.
2. **“Two” fields that are required to be distinct.** Then there is not even a pair of distinct imaginary quadratic fields with the same discriminant, so (E) is false even before units are considered. If $K_1 = K_2$ is allowed, the unit groups are equal.
3. **The class-group condition.** It is one conjunct of an existential, and we refute the existential for an arbitrary predicate in its place. Whatever “3-part” means (Sylow 3-subgroup, 3-rank, Cl/Cl^3), the argument is unaffected. When $K_1 = K_2$ it holds trivially.
4. **Orders and binary quadratic forms.** If “fields of discriminant D ” is read as imaginary quadratic orders of discriminant D , then by (F4) there is only one, and by Theorem 4 its unit group is $\mathbb{Z}/w(D)$, a function of D . If it is read as primitive positive definite binary quadratic forms of discriminant D , the form class group $C(D) \cong \text{Pic}(\mathcal{O}_D)$ is a function of D . The “unit group” of a form f , its group of proper automorphs in $\text{SL}_2(\mathbb{Z})$, is in bijection with the solutions of $t^2 - Du^2 = 4$ via $(t, u) \mapsto \begin{pmatrix} (t - bu)/2 & -cu \\ au & (t + bu)/2 \end{pmatrix}$ (the classical description of automorphs; see Cox, §7, and D. A. Buell, *Binary Quadratic Forms*, Springer 1989, Ch. 3). So it is cyclic of order $w(D)$ for every form of discriminant D . Again nothing depends on more than D .
5. **Absolute value of the discriminant.** Both fields are imaginary, so their discriminants are negative, and $|D_1| = |D_2|$ gives $D_1 = D_2$.
6. **Dropping “of the same discriminant”.** Then pairs with isomorphic 3-parts and non-isomorphic unit groups do exist, but they are tiny. For example $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-2})$ all have class number 1 (trivial 3-parts), with unit groups of orders 6, 4, 2. The discriminants involved are $-3, -4, -8$, with one digit, so clause (M), the seven-digit minimum, fails under this reading too. Moreover, every witness pair must contain $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$, since all other unit groups are $\{\pm 1\}$.

6 The Lean formalization

The directory `lean4/` contains a Lean 4.19.0 project that uses the core library only. All objects are defined from scratch.

- `Squarefree m` ($d^2 \mid m \Rightarrow d = \pm 1$), `IsImagQuadParam m` ($m < 0$ and squarefree), and `disc m`. `SqfCheck` is a decidable test proved sound (`squarefree_of_check`). It gives `param_neg1`, `param_neg2`, `param_neg3`, ...; `not_param_neg4`.
- `disc_injective`, `disc_fundamental` (values $\equiv 1 \pmod{4}$ or $\equiv 8, 12 \pmod{16}$), and `mOfDisc_disc` (explicit inverse): Lemma 1.
- `QR` and `mul t n`: the ring $R_{t,n}$ of pairs. Proven: `mul_comm`, `mul_assoc`, `one_mul`, `mul_one`, `mul_add`, `omega_sq` ($\omega^2 = t\omega - n$), `norm_mul`, `mul_conj`, `four_norm` (equation (1)), and `discBasis_eq` ($\det \text{Tr}(e_i e_j) = t^2 - 4n$).
- `fT m`, `fN m`: the parameters of $\mathcal{O}_m$ in (F2). `sqrtM_sq`: $\sqrt{m} \in \mathcal{O}_m$. `field_discBasis`: the discriminant of $\mathcal{O}_m$ is `disc m`. `halfCoords_mul`: under $a + b\omega \mapsto (x + y\sqrt{m})/2$ the multiplication is that of $\mathbb{Q}(\sqrt{m})$. `integral_iff`: Lemma 3.
- `tD D`, `nD D`: the order of discriminant D , with `discBasis_tD`. `fT_eq`, `fN_eq`: $\mathcal{O}_m$ is the order of discriminant `disc m`.
- `IsUnitQ` (has an inverse), `UnitGrp` (the subtype with its multiplication), and `isUnit_iff_norm`. `units_classification`: for every valid $D < 0$, x is a unit iff $x \in \text{unitList } D$. `unitList.length` ($= w(D)$), `unitList_nodup`, and `unitList_eq_pows`: the list is $1, g, \dots, g^{w-1}$ with $g^w = 1$. Together these say that the unit group is cyclic of order $w(D)$ (Theorem 4).
- `field_units_pm_one`, `field_units_neg1`, `field_units_neg3`, and `field_units_card`: the field case.
- `MulIso`: bijections with inverse that preserve multiplication. `orderUnits_iso_of_w_eq` and `fieldUnits_iso_of_disc_eq`: Corollary 5. The latter does not use `disc_injective`.
- Claim C3: $\exists m_1, m_2$, `IsImagQuadParam m1`, `IsImagQuadParam m2`, `disc m1 = disc m2`, `C3 m1 m2`, and $\neg \text{Nonempty (MulIso (FieldUnits m₁) (FieldUnits m₂))}$. Here C3 is an arbitrary predicate standing for the class-group condition. `conjecture_00000004310_false`: $\neg \text{Claim C3}$ (via injectivity), and `conjecture_00000004310_false'` (via w). `no_witness_discriminant` and `smallest_seven_digit_false` refute clause (M). `order_claim_false` covers the reading with orders.
- Non-vacuity: `fieldUnits_not_iso` proves $\mathcal{O}_{-1}^\times \not\cong \mathcal{O}_{-2}^\times$ by a pigeonhole argument, so the condition “not isomorphic” is not empty. `weakened_reading_small_witness` records the pair $(-1, -2)$ with discriminants $-4, -8$ for reading 6. `hypotheses_satisfiable` and `param_*` show that the field hypotheses are satisfiable.

Polynomial identities are proved by distributing, sorting monomials with `simp only [Int.mul_comm, ...]`, and closing with `omega` (the macro `poly_omega`). The build takes about ten seconds. The `#print axioms` report shows only `propext`, `Quot.sound` and `Classical.choice`. There is no `sorry` and no `native_decide`.

What is not in Lean. Lean does not construct $\mathbb{Q}(\sqrt{m})$ or the ring of all algebraic integers. Facts (F1)–(F4) are the bridge from “imaginary quadratic field” to “squarefree $m < 0$ together with the ring `QR` with parameters `fT m`, `fN m`”. Lean proves the integer part of (F2) (`integral_iff`, `halfCoords_mul`) and the discriminant formula (F3) as the determinant of the trace form. The

rational step $2v \in \mathbb{Z}$ and the uniqueness in (F1) are argued in this report. Class groups are not formalized; they enter only as an arbitrary predicate. Theorem `conjecture_00000004310_false` uses only `disc_injective`; the stronger units classification gives the second proof.

7 Independent verification

`verify.py` (standard library only) checks:

- that `disc` is injective on the 12160 squarefree m with $-20000 < m < 0$; that its values are fundamental discriminants (recognised by an independent test); and that all 6079 negative fundamental discriminants with $|D| < 20000$ arise;
- the unit counts of $\mathcal{O}_m$ by brute force on a box of side 7, giving 4, 6 and 2, so the count is a function of D ;
- for all 4999 discriminants $-10^4 < D < 0$, $D \equiv 0, 1 \pmod{4}$, that $t^2 - Du^2 = 4$ and $N(a+b\omega) = 1$ each have $w(D)$ solutions, and that the unit group is cyclic;
- for binary quadratic forms, that every reduced primitive form of discriminant D (143695 forms for $|D| < 10^4$) has exactly $w(D)$ proper automorphs. The automorphs are found through (t, u) for all D , and by brute force over SL_2 matrices with entries in $[-2, 2]$ for $|D| < 400$;
- a few class numbers, and that no discriminant carries two fields with different unit groups;
- reading 6: $h(-3) = h(-4) = h(-8) = 1$, giving one-digit witnesses.

All checks pass (see `verification.txt`).

8 Conclusion

An imaginary quadratic field is determined by its discriminant, and its unit group is cyclic of order $w(D) \in \{2, 4, 6\}$, a function of the discriminant alone. So no two imaginary quadratic fields of the same discriminant have non-isomorphic unit groups, whatever their class groups are. The same holds for imaginary quadratic orders and for binary quadratic forms. The witness set is empty, so there is no “smallest seven-digit discriminant”. Conjecture 00000004310 is false.